THE EFFECTIVENESS OF GOVERNMENT RESEARCH

Measuring the Scholarly, Judicial, and Regulatory Uptake of Federal-Agency Research

[Author]

DRAFT — FOR DISCUSSION

June 2026

ABSTRACT

Federal agencies are among the largest producers of research in the world. The statistical agencies measure the economy and the population; the Federal Reserve maintains one of the most prolific economics-research enterprises anywhere; the economic-analysis agencies model the effects of their own rules. This study asks a question of governance that the scale of that enterprise invites: how effective is government research—does it, and where does it, actually get used? Using three free, public, reproducible data sources—OpenAlex for scholarly citation, CourtListener for the judicial record, and the Federal Register API for the rulemaking record—it measures the uptake of the research produced by seven federal agencies, grouped into three functional types, across three corpora. Three findings emerge. First, scholarly influence is highly concentrated: the Federal Reserve's research exceeds that of the statistical and economic-analysis agencies combined, and by citation count the Board of Governors ranks among the largest economics-research institutions in the world. Second, that ranking inverts in the courts: the statistical agencies' products—the Consumer Price Index, Bureau of Labor Statistics data—appear in thousands of judicial opinions, while the Federal Reserve's flagship research products appear in almost none. Different kinds of government research prove effective in different arenas, and the institution with the largest research enterprise leaves the smallest footprint where legal consequences are decided. Third, analytical language is a stable, pervasive feature of the rulemaking record across three decades rather than a recent development. The study then specifies, and previews, a lead-lag design to test whether agency research leads policy attention or instead reacts to it, and closes with a candid account of the limits of count-based measurement. All data and code are reproducible.

CONTENTS

Introduction

Few governments produce as much research as the United States, and few of its activities are as seldom examined as a function in their own right. The Census Bureau and the Bureau of Labor Statistics measure the population and the economy; the Federal Reserve maintains research divisions at its Board and twelve Reserve Banks and publishes hundreds of working papers a year; the Securities and Exchange Commission and the Federal Trade Commission operate standing economics units whose studies inform their rules and their cases.[1] Across the government, the production of data, models, and analysis consumes substantial public resources and carries the imprimatur of the state. This study asks a question that the scale of the enterprise invites but that is rarely posed empirically: how effective is government research—whether, and where, it is actually used?

Effectiveness, so framed, cannot be observed directly, and this study does not attempt a causal estimate of the effect of government research on policy outcomes. It measures a tractable proxy—*uptake*: the degree to which the research of the seven agencies, grouped into three functional types (the central bank, the statistical agencies, and the economic-analysis agencies), propagates into three corpora that can be counted—the scholarly literature, the body of judicial opinions, and the rulemaking record. Uptake is an imperfect proxy for influence, and the study is candid about the gap; but the disparities it reveals are sufficiently large, and sufficiently patterned, to support conclusions about where different kinds of government research take hold.

Part I describes the data and method. Part II reports the results across the three corpora. Part III specifies a lead-lag design for the question whether agency research leads policy attention

[1] On the scale of the federal research and statistical enterprise, see Nat'l Acads. of Scis., Eng'g & Med., *Principles and Practices for a Federal Statistical Agency* (7th ed. 2021). This study examines seven agencies in three functional types: the central bank (the Federal Reserve Board and the Reserve Banks of New York and St. Louis), the statistical agencies (the Census Bureau and the Bureau of Labor Statistics), and the economic-analysis agencies (the SEC and the FTC).

or instead follows it. Part IV sets out the threats to inference. A short conclusion draws the findings together.

I. Data and Method

A. Three Corpora, Three Free Sources

The design measures agency-research uptake in three corpora, each through a free public interface that imposes no subscription cost and requires no proprietary access.

Scholarly literature — OpenAlex. OpenAlex is an open bibliographic database that catalogs scholarly works, their authors and institutional affiliations, and the citations among them. For each agency research arm, the study retrieves the institution's catalogued output (works), the cumulative citations to that output, and standard bibliometric summary statistics, including the h-index.[2]

Judicial record — CourtListener. CourtListener, maintained by the Free Law Project, is a comprehensive open database of American judicial opinions. For each distinctive research product, the study counts the opinions whose text contains the product's name, in total and by decade.[3]

Rulemaking record — the Federal Register. The Federal Register API exposes the full text of rules, proposed rules, and notices. For each term, the study counts the documents whose text contains it, in total and by year.[4]

---

[2] OpenAlex, api.openalex.org. Institutions were resolved by name search and confirmed by country and output volume; counts were retrieved through the institutions and works endpoints under OpenAlex's polite-pool conventions (a contact address is supplied with each request). Retrieval date: June 2026.

[3] CourtListener, courtlistener.com, REST API v4. Counts are exact-phrase searches over the opinion corpus; the API returns a total match count and supports date-filtered subqueries by filing date. CourtListener's coverage is broad but not exhaustive, and varies by jurisdiction and era. Retrieval date: June 2026.

[4] Federal Register API, federalregister.gov/developers/api/v1. The API caps reported totals at 10,000; for the most common terms the annual series, which is uncapped at the values observed here, is the reliable measure. Retrieval date: June 2026.

B. Measures and Their Limits

The unit of analysis throughout is the document, and the measure is a count. This simplicity is a virtue for reproducibility and a limitation for precision, and that limitation should be acknowledged at the outset. A count of mentions is not a measure of influence: it cannot distinguish a citation that carries an argument from one that decorates it, and it misses influence that travels without naming its source. Three specific artifacts recur and are flagged wherever they bear on a result.

First, *affiliation attribution*. OpenAlex assigns a work to an institution whenever any contributing author lists that affiliation. This over-counts: the single most-cited work attributed to the Federal Reserve Board in the raw data is a scientific-computing software paper, not economics research, included because a contributor listed a Federal Reserve affiliation.[5] Second, *the Federal Register cap*, which fixes reported totals at ten thousand and makes the annual series the trustworthy measure for common terms. Third, *phrase ambiguity*: a term like "economic analysis" counts all usages, not only those referring to a particular agency's research. None of these artifacts can convert the order-of-magnitude differences reported below into artifacts of measurement, but each counsels against reading small differences as real.

C. Reproducibility

The entire pipeline is scripted and cached. A collection script retrieves the data from the three APIs and stores the raw responses as JSON; a plotting script renders the figures from that cache; the counts reported in the tables are read from the same cache. Re-running the collection

[5] *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python* (2020), which alone accounts for roughly five percent of the Board's catalogued citations. The aggregate counts remain informative as orders of magnitude; individual point estimates should not be over-read.

reproduces the dataset (subject to the corpora having grown since retrieval), and re-running the plotting reproduces every figure exactly. No step depends on a paid service or a credential.[6]

II. Results

A. Scholarly Footprint

By every bibliometric measure, the Federal Reserve is the dominant research producer among the agencies studied. The Board of Governors alone has a catalogued output of more than fourteen thousand works and has accumulated roughly 759,000 citations—more than the Census Bureau, the Bureau of Labor Statistics, the SEC, and the FTC combined—with an h-index in the hundreds that places it among the most-cited research institutions of any kind. Adding the New York and St. Louis Reserve Banks brings the Federal Reserve system's catalogued citations past one million. Table 1 reports the figures; Figure 1 displays the citation comparison.

Table 1. Scholarly footprint of agency research arms (OpenAlex, retrieved June 2026).

Institution (cluster)	Works	Citations	h-index
Federal Reserve Board (Fed)	14,345	759,106	333
FRB New York (Fed)	4,587	295,443	239
U.S. Census Bureau (statistical)	4,722	172,756	182
FRB St. Louis (Fed)	4,202	113,556	138
Bureau of Labor Statistics (statistical)	2,706	91,650	107
SEC (economic-analysis)	1,320	41,478	91
FTC (economic-analysis)	1,610	36,396	90

[6] The accompanying repository contains the data-collection script, a gentle backoff-aware re-fetch for rate-limited endpoints, the figure-plotting script, and the cached JSON responses. The throttling and contact-address conventions are set to honor each service's published use policy.

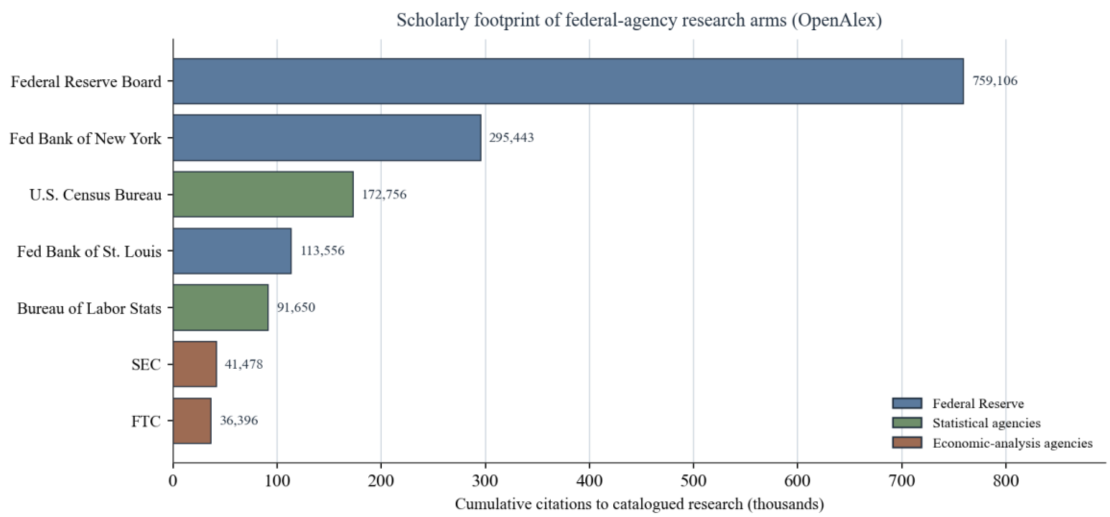

Figure 1. Cumulative scholarly citations to each agency's catalogued research output (OpenAlex). Colors denote the three clusters. Institution-level counts are subject to the affiliation-attribution caveat discussed in Part I.B.

The result is striking on its own terms: the Federal Reserve is not merely a central bank that conducts research but, by the metrics of the scholarly economy, one of the largest economics-research institutions in the world. The disparity is not subtle—the Board's citation count is an order of magnitude larger than the FTC's—and it is robust to the affiliation caveat, which at roughly five percent cannot account for a gap of that magnitude. Measured by scholarly uptake, the central bank's research is by far the most effective; the economic-analysis agencies', the least.

B. Judicial Uptake

In the courts the ranking inverts. The research products that appear most often in judicial opinions are the legally operative statistics and the analysis that conditions rulemaking, not the Federal Reserve's celebrated research. The Consumer Price Index appears in some 3,400 opinions and Bureau of Labor Statistics data in roughly 1,900; "cost-benefit analysis" appears in some 2,700. The Federal Reserve's flagship research products are by comparison invisible: its triennial Survey of Consumer Finances appears in four opinions, the Beige Book in none, and the Senior Loan Officer Opinion Survey in none. Table 2 reports the counts; Figure 2 illustrates the contrast.

Table 2. Judicial opinions mentioning each research product (CourtListener, retrieved June 2026). The 2020s column is partial (through mid-2026).

Product (cluster)	Total	1990s	2000s	2010s	2020s
Consumer Price Index (stat.)	3,397	701	718	812	202
cost-benefit analysis (econ.)	2,725	467	617	775	321
Bureau of Labor Statistics (stat.)	1,932	190	298	476	181
American Community Survey (stat.)	142	0	13	95	34
FTC Bureau of Economics (econ.)	108	10	30	20	2
Survey of Consumer Finances (Fed)	4	0	2	0	1
SEC Div. of Econ. & Risk Analysis (econ.)	4	0	0	0	4
Beige Book (Fed)	0	0	0	0	0
Senior Loan Officer Survey (Fed)	0	0	0	0	0

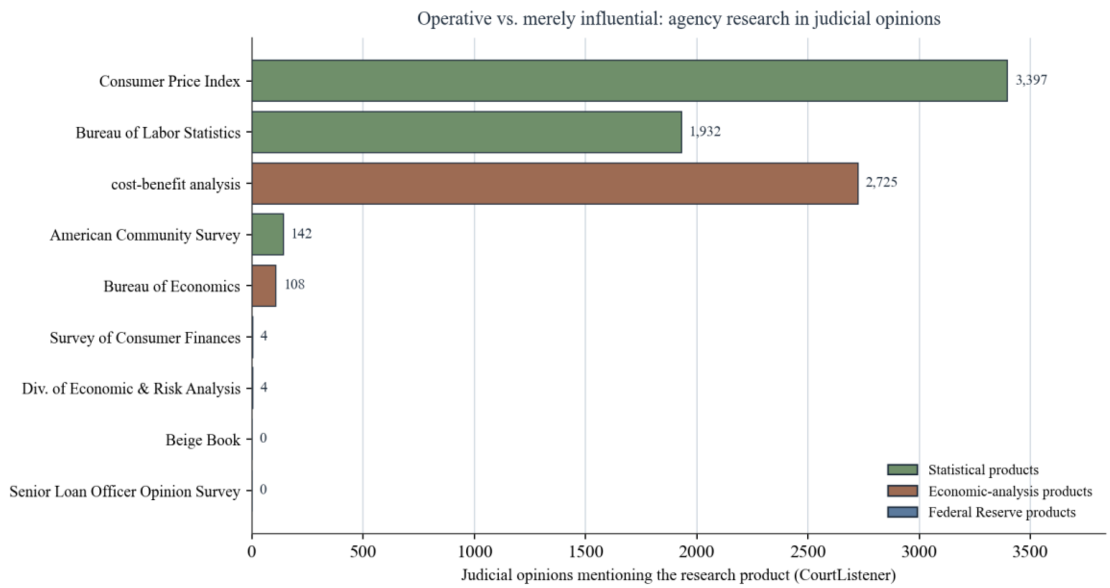

Figure 2. Judicial opinions mentioning each research product (CourtListener). Statistical and economic-analysis products that bear on legal consequences appear thousands of times; the Federal Reserve's flagship research products are nearly absent.

The contrast is the central finding of this study: scholarly influence and legal uptake diverge sharply, and they diverge by type of agency. Research that is legally operative—the statistics incorporated into benefits, taxes, and apportionment, and the analysis tied to reviewable rules—appears where legal consequences are decided, because it helps decide them. Research that is influential only in the academic sense—however vast its scholarly footprint—does not surface in litigation, because it operates through channels the courts do not police. The Federal Reserve has the largest research enterprise and the smallest judicial footprint of the three types; the statistical agencies, more modestly cited in scholarship, dominate the courts. Effectiveness, in other words, is arena-specific: each kind of government research is most used in the forum its outputs are built to serve.

The decade breakdown adds a temporal dimension, shown in Figure 3. The judicial presence of the established statistical products is durable across forty years; the American Community Survey rises from literal nonexistence before its mid-2000s launch to a regular presence in the courts within a decade—a clean natural illustration of a new agency research product passing into regular judicial use.[7]

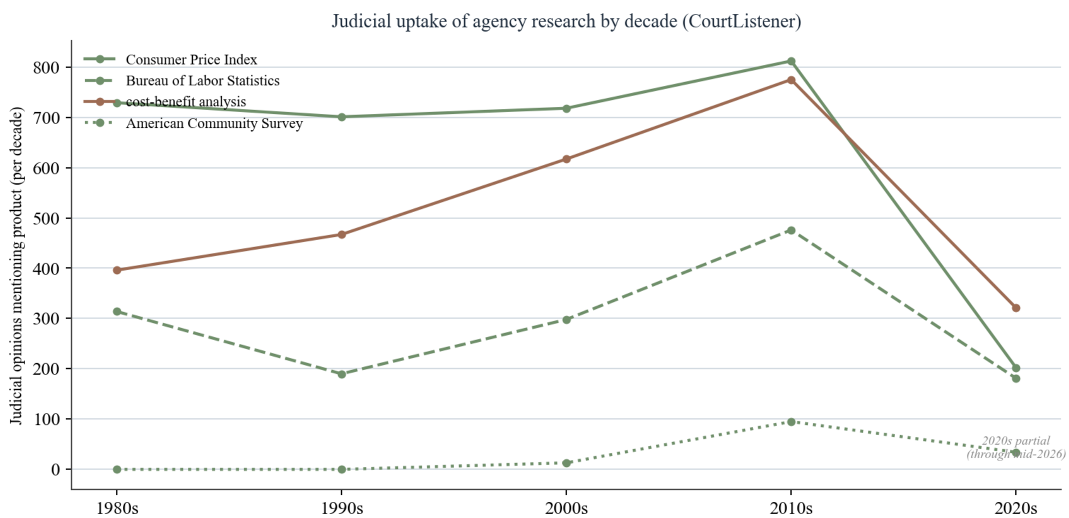

Figure 3. Judicial uptake of agency research products by decade (CourtListener). The 2020s point is partial (through mid-2026). The American Community Survey enters the courts only after its mid-2000s launch.

C. Regulatory Uptake

In the rulemaking record, analytical language is pervasive and, strikingly, stable. The phrase “economic analysis” appears in thousands of Federal Register documents in a typical year, fluctuating roughly between 2,500 and 3,600 annually from 1995 through 2025 without a clear secular trend; “cost-benefit analysis” appears in a smaller but similarly steady band. Specific

[7] The 2020s figures are partial, covering filings through mid-2026, and so understate the full-decade totals; the apparent decline in the last point of each series in Figure 3 is an artifact of the truncated window, not a real reversal.

research products track the salience of their underlying programs: the Consumer Price Index and the American Community Survey appear in hundreds of documents per year, the Survey of Consumer Finances in a handful. Table 3 reports totals and Figure 4 the annual series.

Table 3. Federal Register documents mentioning each term (retrieved June 2026). Totals marked "10,000+" are at the API's reporting cap; for those terms the annual series is the reliable measure.

Term (cluster)	Total (capped at 10,000)	Typical annual count
economic analysis (econ.)	10,000+	~2,500–3,600 / year
Consumer Price Index (stat.)	9,881	~200–500 / year
cost-benefit analysis (econ.)	8,663	~150–450 / year
American Community Survey (stat.)	10,000+	~300–750 / year
Survey of Consumer Finances (Fed)	773	~10–50 / year

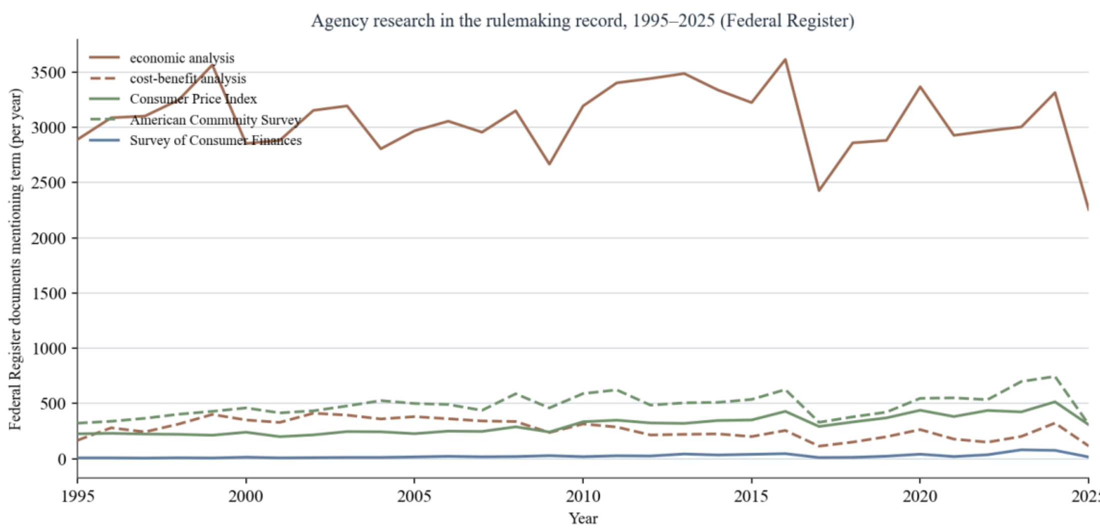

Figure 4. Federal Register documents mentioning selected research terms, by year, 1995–2025. "Economic analysis" is a constant fixture of the rulemaking record; product-specific references are smaller and program-driven.

The stability is the notable result. A common narrative dates the "cost-benefit state" to a series of developments in the 2000s and 2010s—the judicial cost-benefit cases, successive executive orders on regulatory review. The Federal Register data suggest the analytical turn is older and more entrenched: the rulemaking record's appetite for analysis has been roughly constant for three decades. Whatever the courts and the executive orders did, they operated on a regulatory culture in which analysis was already ubiquitous.[8]

III. The Reactive-Literature Hypothesis

The descriptive results establish uptake. They do not establish direction: whether agency research leads the policy and public attention it addresses or follows it. The conventional account assumes leadership—the agency produces knowledge, and policy then acts on it. The reactive-literature hypothesis is the contrary conjecture: that a substantial share of agency research is produced and cited in response to a controversy or a pending decision, rather than in advance of it. The distinction bears directly on what effectiveness means here: research that anticipates policy and research that is mobilized to defend a position already taken are used in very different senses, and only the former supports the familiar case for insulating official research from political control.

The design to test the hypothesis is a cross-correlation, or lead-lag, analysis. For a given research topic, one constructs three time series: the production of agency research on the topic (proxied by the OpenAlex publication-year counts for the relevant works), the policy attention to it (proxied by Federal Register mentions by year), and the judicial attention to it (proxied by CourtListener filings by year). One then estimates the cross-correlation of the research series against each attention series at a range of leads and lags. If research *leads*, the correlation peaks when the research series is shifted earlier in time; if research *lags*—the reactive case—the correlation peaks when the research series is shifted later.

This study implements that design at the topic level rather than reporting estimates here, for a reason the next Part develops: the timing artifacts are severe enough that the estimates should

[8] The flat trend cautions against attributing the prevalence of analysis in rulemaking to any single doctrinal or executive intervention; the analytical character of modern rulemaking appears to predate, and to outlast, the particular cost-benefit cases and executive orders to which it is often credited.

be presented as exploratory, with placebo windows and alternative lag specifications, rather than as a headline finding. What the descriptive data already suggest, and what the lead-lag design is built to test rigorously, is that the conventional account, in which research leads policy, is at best incomplete. The flat trajectory of the Federal Register analysis series, for instance, is hard to square with a story in which research drives a growing analytical demand; it is easier to square with a story in which analysis is produced to satisfy a standing institutional expectation—reactively, on demand.[9]

IV. Threats to Inference

The descriptive findings are robust because they are gross: the gaps they report are of orders of magnitude, and no plausible measurement correction reverses them. Four opinions citing the Survey of Consumer Finances do not become four thousand under any reasonable adjustment; the Board's scholarly dominance survives the removal of mis-attributed works many times over. For the descriptive claims, the threats catalogued below affect precision, not direction.

The causal hypothesis is far more fragile, and candor requires that the principal threats be named. *Publication lag* is the most serious: scholarship responding to a 2020 event may not appear until 2022 or 2023, which can make genuinely reactive research look contemporaneous with, or even anticipatory of, a later event. *Corpus lag* compounds it: the Federal Register and the judicial record have their own institutional delays, so the three series are not on a common clock. *Phrase ambiguity* means the research and attention series may not track the same underlying topic. And

[9] The reactive-literature framing connects to a broader line of inquiry into whether scholarly and official literatures lead or lag the policy developments they address. The present design is a first approach; a fuller treatment would incorporate topic models to define the research and attention series consistently and would exploit discrete policy shocks to sharpen identification.

coverage variation—CourtListener's uneven coverage across jurisdictions and eras, OpenAlex's changing completeness over time—can induce spurious trends.[10]

The honest summary is that the public data are sufficient to establish the central descriptive claim—that government research has large and unevenly distributed downstream effects—and to motivate, but not yet to prove, the reactive-literature hypothesis. That asymmetry is itself a useful result: the case for treating the effectiveness of government research as a serious empirical question does not depend on resolving the harder matter of direction, and can be made on the counts alone.

Conclusion

Three free databases, interrogated with nothing more elaborate than counts, suffice to answer the study's guiding question with some precision: government research is effective, but its effectiveness is uneven and arena-specific. The Federal Reserve's research is the most prolific and most-cited in the federal government, by a substantial margin—and almost entirely absent from the judicial record, where the statistical agencies' legally operative numbers predominate. Analytical language pervades the rulemaking record and has done so steadily for thirty years. Each kind of government research takes hold in the forum its outputs are built to serve, and the institution with the largest research enterprise leaves the faintest trace in the corpus where legal consequences are determined. The harder question—whether agency research leads policy attention or instead reacts to it—remains open, and the lead-lag design set out above is a first attempt at it rather than a settled answer. But the descriptive case is already made: the research function of the federal

[10] The study's mitigations include: restricting lead-lag estimates to topics with well-defined, unambiguous product names; using placebo event windows to gauge the false-positive rate; reporting results under multiple lag conventions; and treating the earliest years of each corpus, where coverage is thinnest, with caution. Even so, the causal estimates are offered as exploratory.

government is large, consequential, and unevenly used, and it rewards the kind of measurement undertaken here.